

Proposal for MLOPS Project

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Project Title	Swiss Segmented Unemployment Rate Predictor
Repository	https://github.com/theodorahaimoff/HSLU_HS26_MLOPS
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1 Problem Statement

1.1 What & for whom

The system predicts Switzerland’s ILO unemployment rate for the *next quarter*. It computes this forecast independently for the national total and for each of 44 demographic segments. These segments come from gender (male or female), nationality (Swiss or foreign), and age bracket (15–24, 25–39, 40–54, or 55–64), taken individually or in every combination, for 45 series in total including the national total. All 45 series are forecast automatically on every run. A checkbox UI lets the user view any one of the 45 already-computed forecasts, but the prediction itself is not user-selected: every segment is always predicted. Target users are social-policy researchers, NGOs, and job seekers who want a near-term outlook broken down by demographic group, not only the single national headline figure.

1.2 Horizon & scope

The forecast horizon is one quarter ahead, roughly three months, for Switzerland. It uses data from 2010–Q1 onward, the earliest quarter at which all 45 segment combinations are consistently published. The 65+ bracket is excluded, since it falls outside standard working age.

1.3 Success criterion

The evaluation metric is Mean Absolute Error (MAE) on quarter-ahead predictions, per segment, compared against a seasonal-naive baseline, defined as “next quarter equals the same quarter last year.” The target is to beat this baseline by at least 15% relative MAE; this figure is a stated goal, not yet confirmed against the data, and will be revisited once initial backtest results are available in MS2/MS3. This is evaluated with a walk-forward backtest: the model is retrained and re-evaluated at each of the last 12 quarterly cutoffs (the last 3 years), and the reported MAE is the average across these 12 evaluation points, per segment.

2 Originality & Motivation

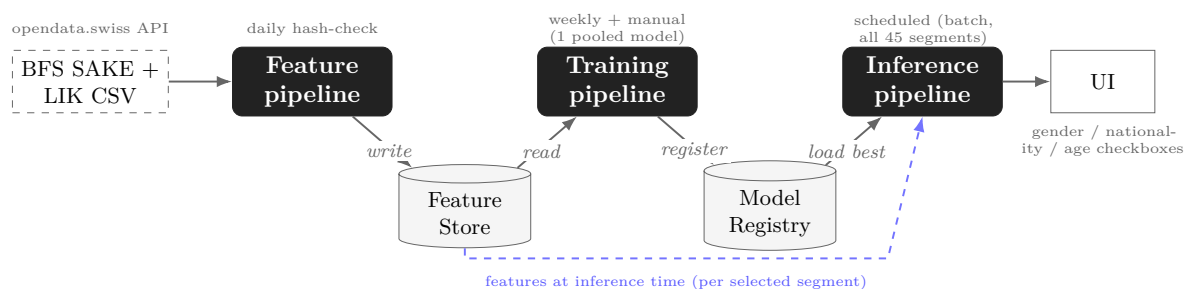
The unemployment rate is one of the clearest indicators of how well a country’s economy is functioning. A breakdown by demographic group is especially informative for marginalized populations. A rising or persistently higher rate for a specific gender, nationality, or age group is often a direct signal of financial hardship that a single national figure hides entirely. Switzerland’s own statistics already let these breakdowns be browsed, but only as a historical record, one group at a time, always looking backward. What is missing is a system that turns this data into a forward-looking signal, at the national level and for any specific demographic segment, rather than only showing what already happened. This project’s twist is *granular, multi-segment forecasting*. The training pipeline produces 45 independent one-quarter-ahead forecasts every cycle, covering the national total plus every demographic breakdown (gender $\times$ nationality $\times$ age), so that emerging hardship in a specific group can be anticipated a quarter ahead rather than only confirmed after the fact. The model also incorporates consumer-price inflation as a covariate, since unemployment and inflation are classically linked through the Phillips curve, and rising inflation can itself be an early signal of broader economic strain.

3 Data Source & Features

Source & provider	Federal Statistical Office (BFS/OFS), Swiss Labour Force Survey (SAKE), ILO-defined unemployment rate. Accessed via the <code>opendata.swiss</code> CKAN API (<code>package_show</code>), which resolves to a stable CSV resource at <code>dam-api.bfs.admin.ch/hub/api/dam/assets/{id}/master</code> . No scraping; a documented public API.
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Secondary source	Federal Statistical Office (BFS/OFS), Swiss Consumer Price Index (Landesindex der Konsumentenpreise, LIK), also via the <code>opendata.swiss</code> CKAN API. A single national series (no demographic breakdown), joined onto every segment's row for the matching quarter as a shared macroeconomic covariate.
Update frequency	The SAKE survey is released quarterly; the LIK index is released monthly and averaged to quarterly to match. The feature pipeline runs daily (GitHub Actions cron) and compares each resource's <code>hash</code> field against the last-seen value to detect a new release; a CSV is re-fetched only when its hash changes.
Data volume & growth	66 quarters (2010–Q1 to 2026–Q2) $\times$ 45 segments are already published at project start; the series grows by one point per segment per quarter. History is bounded by BFS's own publication start date for this breakdown, not by when this project's poller starts.
Label	The officially published unemployment rate for quarter $T + 1$ for the selected segment — an independently surveyed value, not derivable from any engineered feature.
Features (named)	Lagged unemployment rates at $T, T - 1, T - 2, T - 4$ (1-year seasonal lag); a rolling 4-quarter mean; quarter-of-year (one-hot); one-hot encodings for gender, nationality, and age bracket; CPI level and quarter-on-quarter inflation rate as a shared macroeconomic covariate.
Rare positive class	Not applicable — regression task, no class imbalance.
Leakage & split	Strictly chronological, walk-forward validation: at each of the last 12 quarterly cutoffs, the model is trained only on data up to that cutoff and evaluated on the following quarter. No cutoff ever trains on data from its own evaluation quarter or later, and rows are never shuffled, since lag features would otherwise leak future information into training rows.

4 System Design



4.1 Tech stack

- **Model:** a single pooled LightGBM gradient-boosted tree model, trained across all 45 segments with segment identity and CPI as input features. Gradient-boosted trees are a standard choice for grouped, tabular time-series forecasting of this kind, handle the mix of lagged numeric and categorical segment features directly, and train in seconds on a dataset this size.
- **Ingestion & scheduling:** GitHub Actions cron. This is provided at no cost, integrates directly with the repository, and is sufficient for a daily hash check across both sources.
- **Feature store:** versioned Parquet files on GCS. A lightweight, file-based feature store is sufficient given the dataset size, so a managed deployment of Hopsworks or Feast is not required.
- **Experiment tracking:** Weights & Biases (W&B). Runs are logged to a hosted dashboard, so no separate tracking server needs to be deployed or maintained. The free academic tier covers this project's storage and usage requirements.
- **Model registry:** W&B Model Registry. It is co-located with experiment tracking under the same account, which avoids introducing a second tool, and supports versioning and stage promotion of the pooled model.
- **Training trigger:** scheduled (weekly) plus manual dispatch, retraining the pooled model described above.
- **Serving:** a containerized FastAPI backend and Streamlit frontend, deployed on Google Cloud Run.
- **Monitoring:** on each new BFS release, the prior quarter's prediction is compared against the now-published actual value per segment, and the resulting error is logged to flag drift.